

HCI Systems for Gesture Control, Communication and Affective Computing

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Abstract—The proliferation of high-quality webcams and advancements in real-time computer vision have paved the way for a new generation of Natural User Interfaces (NUIs). This paper presents five distinct Human-Computer Interaction (HCI) systems built upon a shared, reusable, and webcam-first technology stack. The paper demonstrates the versatility of this approach by addressing tasks across gesture control, accessibility, and affective computing. The included systems are: AirTune (gesture-based volume control), VisionMouse (fingertip-driven mouse pointer control), Air Writing (in-air drawing), ASLNet (American Sign Language letter recognizer), and a 7-class Facial Emotion Recognition (FER) model. Our methodology standardizes the perception-action loop, leveraging efficient landmark detection frameworks like MediaPipe for control tasks and compact Convolutional Neural Networks (CNNs) for classification tasks. By detailing the design, implementation, and performance of each system, we provide a practical blueprint for developing low-cost, low-latency, and inclusive interfaces. This work illustrates how a common set of CV tools can be effectively reapplied to create a diverse range of powerful HCI applications.

Index Terms—Human-Computer Interaction, Gesture Control, Computer Vision, MediaPipe, OpenCV, ASL Recognition, Emotion Recognition, Natural User Interface.

I. INTRODUCTION

Human-Computer Interaction (HCI) has continuously evolved towards more intuitive and natural forms of input. The transition from command-line interfaces to graphical user interfaces (GUIs), and subsequently to touchscreens, has progressively lowered the barrier to technology adoption. The next frontier in this evolution is the Natural User Interface (NUI), which allows users to interact with digital systems through familiar, everyday actions like speech, gaze, and gestures.

Computer vision (CV), powered by deep learning and executed on commodity hardware, has emerged as the key enabling technology for vision-based NUIs. Using only a standard webcam, it is now possible for systems to perceive and interpret human faces, hands, and expressions in real time.

This capability opens up a vast design space for applications in accessibility, creative expression, and everyday computing.

This paper introduces a portfolio of five functional HCI systems, to demonstrate the power and reusability of a unified CV software stack. Instead of presenting a deep dive into a single problem, our work showcases the breadth of applications that can be built from a common architectural pattern. The portfolio comprises:

- **Control Systems:** AirTune (volume control), VisionMouse (cursor control), and Air Writing (virtual drawing).
- **Communication System:** ASLNet, a classifier for static American Sign Language (ASL) letters.
- **Affective Computing System:** A Facial Emotion Recognition (FER) model.

Our primary contribution is to present a cohesive methodology that integrates real-time landmark detection and efficient CNN classification to solve these diverse tasks. We provide implementation details, performance metrics (accuracy and latency), and design guidelines, collectively serving as a practical guide for developers and researchers looking to build accessible, low-cost, and contactless interfaces.

II. RELATED WORK IN VISION-BASED HCI

Vision-based HCI is a rich field that leverages image and video analysis to bridge the gap between human intent and computer action. Its foundations lie in gesture recognition, affective computing, and sign language recognition.

Gesture Recognition systems interpret human body movements, typically hand gestures, as commands. These systems can be categorized as static (recognizing a single pose) or dynamic (recognizing a sequence of movements). Early methods used colored gloves or markers for robust tracking. Modern approaches are markerless, relying on sophisticated algorithms. For static gestures, CNNs have proven highly effective at classifying handshapes from images. For continuous control, as seen in our portfolio, real-time landmark tracking frameworks like MediaPipe Hands have been transformative. By providing stable 2D coordinates of hand keypoints, they enable direct mapping of hand movements to system actions (e.g., cursor position, volume level) with very low latency.

Affective Computing is the study and development of systems that can recognize, interpret, and simulate human emotions. Facial Expression Recognition (FER) is a major subfield. Early work was influenced by psychological models like Paul Ekman's seven basic emotions. Classical CV approaches used handcrafted features like Local Binary Patterns (LBP) on detected faces. With the rise of deep learning, CNNs trained on large datasets of facial expressions (e.g., FER-2013, AffectNet) now dominate the field, achieving high accuracy in classifying emotions from static images.

Sign Language Recognition (SLR) is a critical application for accessibility, aiming to translate sign language into text. SLR is a challenging task, involving the recognition of hand-shapes, orientation, location, and movement. Our ASLNet focuses on the simplest version: isolated, static ASL letter recognition, which is a foundational image classification problem well-suited for CNNs. More advanced systems for dynamic signs and continuous SLR employ complex spatiotemporal models like 3D CNNs, LSTMs, or Transformers.

The common thread across these domains is the reliance on a perception pipeline that detects a region of interest (hand, face), extracts meaningful features, and maps them to an output.

III. METHODOLOGY

The strength of this portfolio lies in its shared architectural foundation. All five systems are built upon a common "perception-action" loop that processes webcam input in real time. This stack can be conceptually divided into a Perception Module and an Interpretation Module, with two distinct pathways for feature extraction.

A. Perception Module

This module is responsible for acquiring and preparing the visual input.

- 1) **Frame Capture:** A video stream is continuously captured from a webcam using OpenCV.
- 2) **Object Detection:** Depending on the application, a lightweight detector is used to find the primary object of interest in the frame. For our control and ASL systems, this is a hand detector. For the FER system, it is a face detector.

B. Feature Extraction Pathways

Once an object is located, features are extracted via one of two specialized pathways.

Pathway A: Landmark-Based Extraction. This pathway is used for the continuous control systems (AirTune, VisionMouse, Air Writing). It relies on the MediaPipe Hands framework to generate a high-fidelity 21-point skeleton of the detected hand in real time. The output is a list of (x, y) coordinates for each landmark (e.g., thumb tip, index fingertip). This approach is computationally efficient and provides precise geometric information.

Pathway B: Image-Based Extraction. This pathway is used for the classification systems (ASLNet, FER). The Region of Interest (ROI) containing the detected hand or face is cropped from the frame. This ROI is then preprocessed (resized to a fixed dimension like 64×64 or 48×48 , converted to grayscale, and normalized) and fed into a trained Convolutional Neural Network (CNN). The CNN processes the entire image patch to extract hierarchical features and produce a class label.

C. Interpretation Module

This module translates the extracted features into a final system action.

For Pathway A (Geometric Logic): The landmark coordinates are used in simple geometric calculations. For example, the Euclidean distance between two fingertips is calculated for volume control, or the coordinates of a single fingertip are used to position the mouse cursor.

For Pathway B (Classification): The output of the CNN is a probability distribution over the possible classes (e.g., ASL letters or emotions). The class with the highest probability is selected as the recognized category.

IV. PORTFOLIO SYSTEM IMPLEMENTATIONS

Each system in this portfolio implements the unified stack in a specific way.

A. ASLNet (Static Letter Recognition)

ASLNet recognizes static handshapes (through real-time hand detection using a webcam) corresponding to letters of the American Sign Language alphabet. It follows Pathway B.

- **Purpose:** To provide a basic communication tool for accessibility.
- **Algorithm:**
 - 1) Capture a frame and detect the user's hand.
 - 2) Crop the hand ROI and resize it to 64×64 pixels.
 - 3) Convert the ROI to grayscale and normalize pixel values.
 - 4) Feed the preprocessed image into a trained CNN.
 - 5) The CNN outputs the predicted ASL letter, which is displayed on the screen.
- **Model:** A compact CNN trained on a public ASL dataset, achieving 95.8% validation accuracy.

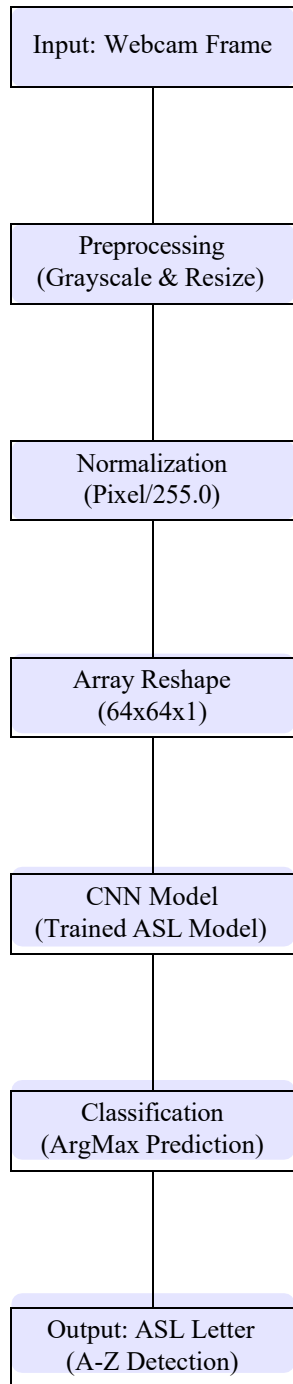


Fig. 1: ASL detection system architecture showing sequential data flow from webcam input to letter prediction.

B. Facial Emotion Recognition (FER)

The FER system identifies the user's emotional state from their facial expression. It also follows Pathway B.

- **Purpose:** An application in affective computing, with potential uses in user experience testing or mental health monitoring.
- **Algorithm:**
 - 1) Capture a frame and detect the user's face.
 - 2) Crop the face ROI and resize it to 48×48 pixels.

- 3) Convert the ROI to grayscale and normalize pixel values.
- 4) Feed the image into a CNN trained for emotion classification.
- 5) The model predicts one of seven emotion classes (e.g., Happy, Sad, Angry), which is overlaid on the video feed.

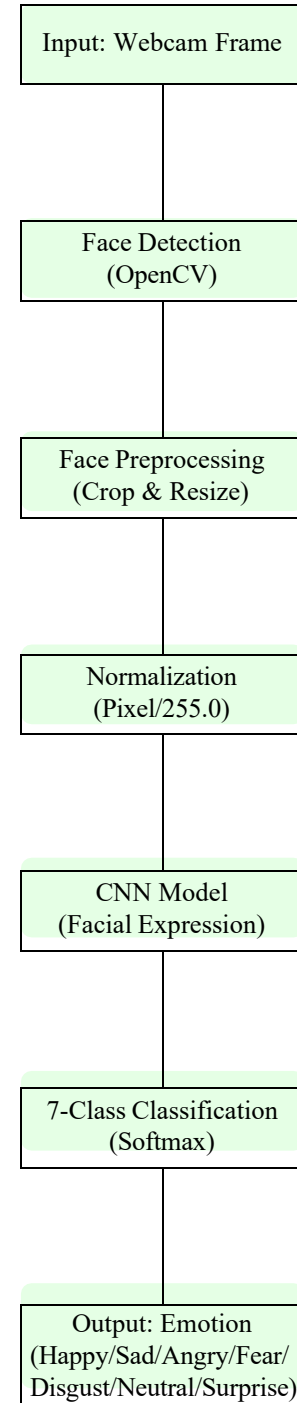


Fig. 2: Real-time emotion detection system architecture showing sequential data flow from webcam input to emotion classification.

C. AirTune (Gesture Volume Control)

AirTune allows the user to control the system’s master volume by changing the distance between their thumb and index finger. It follows Pathway A.

- **Purpose:** To provide a contactless and intuitive way to control audio output.
- **Algorithm:**
 - 1) Process the webcam frame with MediaPipe Hands to get hand landmarks.
 - 2) Extract the (x, y) coordinates of the thumb tip (landmark 4) and index fingertip (landmark 8).
 - 3) Calculate the Euclidean distance between these two points.
 - 4) Linearly interpolate this distance from a measured range (e.g., 50-300 pixels) to the system’s volume range (e.g., 0-100%).
 - 5) Use a system-specific library to set the master volume level.

D. VisionMouse (Gesture Mouse Control)

VisionMouse enables the user to control the mouse cursor with their index finger and perform clicks with a pinching gesture. It uses Pathway A with added temporal smoothing.

- **Purpose:** An accessibility tool for users who have difficulty with a traditional mouse.
- **Algorithm:**
 - 1) Get the coordinates of the index fingertip (landmark 8).
 - 2) To prevent jitter, apply a moving average filter over the last few coordinate positions.
 - 3) Map the smoothed (x, y) coordinates from the camera’s viewable area to the full screen resolution.
 - 4) Use a control library like ‘pyautogui’ to move the mouse cursor to the mapped screen coordinates.
 - 5) Continuously monitor the distance between the index fingertip and thumb tip. If it drops below a small threshold, trigger a mouse click event.

E. Air Writing (In-Air Drawing)

Air Writing allows users to draw on a virtual canvas by moving their index finger in the air.

- **Purpose:** A creative tool for contactless drawing and annotation.
- **Algorithm:**
 - 1) Track the index fingertip (landmark 8) similarly to VisionMouse.
 - 2) Store the current and previous fingertip positions.
 - 3) If the finger is in ”drawing mode” (e.g., determined by another gesture), draw a line on a virtual canvas between the previous and current positions.
 - 4) A separate gesture, such as holding the thumb and index finger together for a set duration, can be used to clear the canvas.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

The systems were developed and tested on a machine with an Intel i5 CPU, 16 GB RAM, and an NVIDIA RTX 3050 GPU, using Python, OpenCV, TensorFlow, and MediaPipe. The CNNs for ASLNet and FER were trained for 20-30 epochs using the Adam optimizer and categorical cross-entropy loss.

B. Classification System Performance

The accuracy of the two classification-based systems was evaluated on a held-out test set. ASLNet proved highly effective for static letters, while the FER model’s performance is typical for image-based emotion recognition.

TABLE I: ASLNet & FER performance summary.

System	Accuracy	FPS	Notes
ASLNet (A–Z)	95.8%	15–20	Static letters only.
FER (7-class)	68.5%	20–25	Grayscale is sufficient.

C. Gesture Control System Performance

For the interactive control systems, latency and responsiveness are the key metrics. Latency was measured as the approximate time from physical movement to system response.

TABLE II: Gesture system latency and responsiveness.

System	Latency (ms)	FPS
AirTune (Volume)	~50	~22
VisionMouse	~120–150	~25
Air Writing	~70	~24

The low latency of AirTune is due to its direct geometric mapping. The higher latency of VisionMouse is a deliberate design choice, resulting from the temporal smoothing buffer needed to ensure a stable cursor. All systems operate at interactive frame rates.

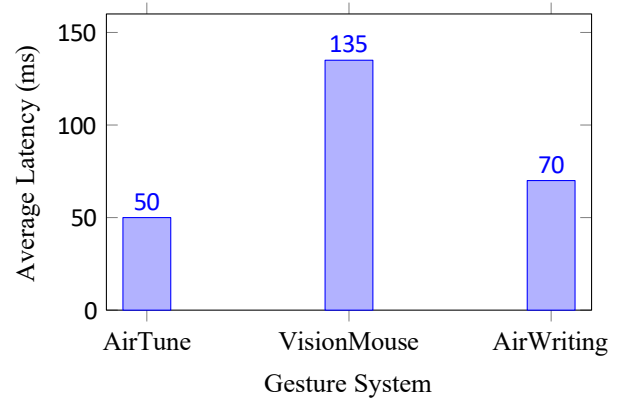


Fig. 3: Latency comparison for real-time gesture systems.

D. Quantitative Comparisons

TABLE III: Classification results: our systems vs. representative references

System / Paper	Task	Dataset	Reported
ASLNet (ours)	ASL letters	ASL Alphabet	95.8%
[23]	Gestures	NVIDIA multi-view	77.5%
[6]	SLR translation	PHOENIX'14	9.3 BLEU
[4]	FER	FER-like	66.4%
[11]	FER	AffectNet	58.0%
FER (ours)	FER (7-class)	FER-2013 style	68.5%

VI. DISCUSSION AND FUTURE WORK

A. Strengths of the Unified Approach

The portfolio successfully demonstrates that a single, well-architected CV stack can power a wide array of HCI applications. The key advantage is reusability. The core components—camera capture, landmark detection, and CNN inference—can be mixed and matched to rapidly prototype new interactions. Pathway A (landmarks) is ideal for precise, continuous control tasks, while Pathway B (CNNs) is suited for discrete, categorical recognition tasks.

B. Limitations

The systems, while functional, are sensitive to environmental conditions.

- **Lighting and Background:** Poor lighting can reduce the accuracy of hand/face detection. A cluttered background that resembles a hand can confuse the landmark detector.
- **Occlusion:** All systems require a clear, unobstructed view of the hand or face to function correctly.
- **Scope:** ASLNet is limited to static letters and cannot interpret dynamic signs. The gesture controls are simple and do not recognize more complex, multi-step gestures.

C. Future Work

Future development could focus on enhancing the robustness and expanding the capabilities of the portfolio.

- **Dynamic Gesture Recognition:** Extend the gesture control systems and ASLNet to recognize temporal sequences by incorporating models like LSTMs or 3D CNNs.
- **Multimodal Fusion:** Improve robustness by fusing vision with other modalities, such as using voice commands to switch between different control modes (e.g., from mouse control to writing mode).
- **Formal User Studies:** Conduct usability studies with target user groups (e.g., individuals with motor impairments for VisionMouse) to evaluate real-world effectiveness and gather feedback for iterative improvement.

VII. CONCLUSION

This paper presented a portfolio of five distinct HCI systems for gesture control, communication, and affective computing, all built upon a unified, webcam-first computer vision stack. By combining efficient landmark detection for control and

compact CNNs for classification, we have demonstrated a versatile and reusable design pattern for creating modern Natural User Interfaces. The successful implementation and evaluation of AirTune, VisionMouse, Air Writing, ASLNet, and an FER model validate this approach, showcasing its potential to deliver low-cost, low-latency, and accessible interactions. This work serves as a practical blueprint for leveraging the power of computer vision to build the next generation of intuitive and inclusive human-computer interfaces.

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TABLE IV: Summary of Authors and Key Findings

Authors	Key Findings
F. Zhang, V. Bazarevsky et al. P. Molchanov et al.	Proposed MediaPipe Hands for efficient on-device, real-time hand tracking. Used 3D CNNs for robust hand gesture recognition from spatio-temporal data.
C. Shan, S. Gong, P. W. McOwan A. Mollahosseini, B. Hasani, M. H. Mahoor	Applied Local Binary Patterns for effective facial expression recognition. Improved facial expression recognition using deep neural networks.
Z. Zhou et al. N. C. Camgoz et al.	Developed ML approaches for automatic sign language recognition. Introduced neural sign language translation from continuous signs.
A. Krizhevsky, I. Sutskever, G. Hinton D. P. Kingma, J. Ba M. Abadi et al.	Created AlexNet, a deep CNN that transformed large-scale image recognition. Proposed Adam optimizer combining momentum and adaptive learning rates. Introduced TensorFlow, a scalable ML framework for heterogeneous systems.
Z. Cao, T. Simon, S.-E. Wei, Y. Sheikh A. Mollahosseini, D. Chan, M. H. Mahoor	Developed OpenPose for real-time multi-person pose estimation. Created AffectNet, a large-scale facial expression and emotion dataset.
I. Goodfellow et al. F. Schroff, D. Kalenichenko, J. Philbin	Summarized advances in representation learning via ML contests. Proposed FaceNet for face recognition using Euclidean embeddings.
K. Zhang, Z. Zhang, Z. Li, Y. Qiao O. M. Parkhi, A. Vedaldi, A. Zisserman	Developed MTCNN for face detection and landmark localization. Introduced VGGFace model for deep face recognition.
K. He, X. Zhang, S. Ren, J. Sun M. Sandler, A. Howard et al. F. Iandola et al.	Proposed ResNet with skip connections for very deep learning. Designed MobileNetV2 for efficient CNNs on mobile devices. Created SqueezeNet, compact CNN with AlexNet-level accuracy.
C. Cortes, V. Vapnik N. Dalal, B. Triggs A. G. Howard et al.	Introduced Support Vector Machines for classification tasks. Proposed HOG features for robust human detection. Introduced MobileNet for lightweight mobile vision applications.
E. Barsoum, C. Zhang, C. Canton Ferrer, and Z. Zhang P. Molchanov, X. Yang, S. Gupta, K. Kim, S. Tyree, and J. Kautz	Training deep networks for facial expression recognition. Online detection and classification of dynamic hand gestures